

# SIFT-Sentinel Findings Report

**Case: BASE-DC intrusion triage**

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Evidence root: /mnt/cases

Audit log: audit/execution-log.jsonl

## SIFT-Sentinel Triage Report

Host: base-dc.shieldbase.lan (Windows Server 2016 domain controller) Evidence root: /mnt/cases

Memory image: /evidence/base-dc-memory.img Case notes / IOCs supplied: none ? unbiased full-host sweep

## Tooling Limitations Affecting Confidence

Read these first; they cap the confidence of several findings below.

? The memory image returned 0 records for both mem\_pslist (call-000009) and mem\_netscan (call-000010), with no error. Live-process and C2 analysis is unavailable: injected processes and external connections can be neither confirmed nor excluded.

? Prefetch is absent (call-000008) ? disabled on this DC. The primary cross-corroborator for evidence of execution is gone.

? parse\_event\_logs exposes only event\_id, time, computer, and description ? no account name, source IP, logon type, or service name/ImagePath. Logon and service-install attribution is therefore impossible with current tooling, capping several items at UNCERTAIN.

## Timeline of Suspicious / Notable Activity

? 2018-05-08 22:09 ? McAfee Agent install (mfemactl.exe in Administrator Temp; Agent dirs). Benign. (call-000007, call-000013, call-000014)

? 2018-08-16 to 2018-09-07 ? Recurring interactive PowerShell by rsydow-a on the DC; transcripts on 08-16 (x3), 08-17, 08-28, 09-03, 09-07. (call-000015)

? 2018-09-04 13:11 to 2018-09-05 22:56 ? Failed-logon storm, ~15-minute cadence, 204 events. (call-000012)

? 2018-09-05 12:16 / 12:27 ? Windows\Temp\DSWBFB8.tmp, DSW5BA0.tmp created. (call-000014)

? 2018-09-06 22:11 (x2) ? Clustered service installs, off the normal 2-day cadence. (call-000013)

? 2018-09-07 20:25:57 ? Windows\NTDS\temp.edb created (AD database temp). (call-000014)

? 2018-09-07 20:26 to 20:30 (x2) ? Clustered service installs. (call-000013)

? 2018-09-07 20:29 to 21:00 ? rsydow-a interactive session: GPP Drives.xml, PowerShell transcript (16:36 local), IE history, Cortana/Media caches. (call-000015)

## Confirmed Findings [CONFIRMED]

None meet the two-independent-source bar. Prefetch is gone and the memory image is empty, so the usual corroboration chains (Amcache + Prefetch + MFT) cannot be completed. That degradation of corroboration capacity is itself a finding.

## Inferred Findings [INFERRED]

? Host is substantially a clean DC baseline. Amcache (call-000006, 37 records) is almost entirely signed OS binaries; the only user-writable-path executables are benign installers (wdksetup.exe = Windows Driver Kit, mfemactl.exe = McAfee). The MFT Temp sweep (call-000014, 913 records) found zero deleted executables, zero malware-like binaries, and no double-extensions or ADS in Temp. Confirmed by: a working memory image plus Prefetch would allow ruling out memory-only / fileless implants.

? Automated failed-logon storm, 2018-09-04 13:11 to 2018-09-05 22:56, metronomic ~15-minute cadence with triple-bursts (call-000012). The mechanical regularity points to an automated agent or scheduled task with stale credentials, not interactive brute force. Confirmed by: the account name and source IP from raw 4625 EventData, which current tooling does not expose.

? rsydow-a ran PowerShell interactively on the DC repeatedly (08-16 through 09-07) with transcription enabled (call-000015). The -a suffix implies an admin account; consistent with either legitimate DC administration or hands-on-keyboard activity. Confirmed by: reading the transcript contents via a sanctioned read path.

## **Contradictions [CONTRADICTION]**

? None surfaced between sources. The only tension is absence vs. expectation: a DC with 26,811 successful logons and an active admin produced an empty memory capture, which is inconsistent with a live host. Treat /evidence/base-dc-memory.img as missing / unreadable / wrong path, not as "no activity."

## **Items Needing Follow-up [UNCERTAIN]**

? Windows\NTDS\temp.edb created 2018-09-07 20:25:57 immediately before two 7045 service installs (20:26 to 20:30). The pattern matches NTDS.dit / VSS credential-extraction TTPs (ntdsutil IFM), but is equally consistent with routine AD maintenance or reboot.

Unattributable and uncorroborated; cannot escalate.

? Off-cadence 7045 clusters on 09-06 22:11 and 09-07 20:26 break the otherwise every-2-days update rhythm, but without service names (tool limitation) their nature is unknown.

? Windows\Temp\DSW\*.tmp (09-05): unidentified temp files; not executables.

## **MITRE ATT&CK Mapping (candidate, evidence-limited)**

? T1110 Brute Force / Credential Stuffing ? failed-logon storm (call-000012). Low confidence; could be a misconfigured service account.

? T1059.001 Command and Scripting Interpreter: PowerShell ? recurring transcripts for rsydow-a (call-000015).

? T1003.003 OS Credential Dumping: NTDS ? hypothesis only, from NTDS\temp.edb plus service-install timing (call-000014, call-000013). Not corroborated.

? T1543.003 Create or Modify System Process: Windows Service ? off-cadence 7045 installs (call-000013). Nature unknown.

## **Recommended Next Steps (prioritised)**

1. Fix the memory evidence. Verify the path/mount of /evidence/base-dc-memory.img and re-run mem\_pslist and mem\_netscan. An empty DC capture is a collection failure, not a result.
2. Pull raw 4625 / 4624 EventData (TargetUserName, IPAddress, LogonType, WorkstationName) to attribute the failed-logon storm ? the single most informative gap.

3. Read the rsydow-a PowerShell transcripts (especially ...20180907163638.txt and the 08-16 set) via the audited read path; they likely contain verbatim commands run on the DC.
4. Retrieve 7045 service names / ImagePaths for the 09-06 22:11 and 09-07 20:26 clusters; correlate against the NTDS\temp.edb write to confirm or deny NTDS.dit extraction.
5. Add an execution corroborator absent Prefetch: pull SRUM, ShimCache (SYSTEM hive), and the full-detail Amcache set to rebuild the execution timeline.

## Bottom Line

No malware, dropped binaries, deleted executables, or C2 were found in the available artifacts. The host looks largely like a clean DC baseline. The three real leads ? an automated failed-logon storm, repeated admin PowerShell by rsydow-a, and NTDS/temp plus off-cadence service activity on 09-06/07 ? are all currently unattributable because the memory image is empty and the event-log tool omits account/IP/service fields. Resolve those two gaps before drawing conclusions.

## Chain-of-Custody Integrity Check

PASS: all 9 cited call\_id(s) resolve to a logged tool invocation in the audit trail.

Note: 6 logged call(s) are not cited in the findings (informational, not an error).

## Audit Trail

16 tool invocation(s), append-only. Every finding above cites a call\_id resolved here.

```
call-000003 analyze_prefetch [ERROR] 0 ms 2026-06-14T13:42:15Z
  args: prefetch_path=/mnt/cases/Windows/Prefetch/
  error: No .pf files found in /mnt/cases/Windows/Prefetch ? Prefetch may be disabled (common on
Windows Server/DC).
call-000001 get_amcache [exit=0] 6390 ms 2026-06-14T14:05:37Z
  binary: AmcacheParser
  sha256: bcf639115a6b5995ef56734c76d3d40fd8b758b687a111b2a6f6744d91b9a78d
  args: amcache_hive=/mnt/cases/Windows/appcompat/Programs/Amcache.hve
call-000002 extract_mft_timeline [exit=0] 83601 ms 2026-06-14T14:06:03Z
  binary: MFTECmd
  sha256: 4b829ec61b34a9817d8eb44ae36e9dfc2e90fdc8f429eec40127ff34cc2f18da
  args: mft_file=/mnt/cases/$MFT, path_filter=None
call-000001 extract_mft_timeline [exit=0] 58719 ms 2026-06-14T14:09:06Z
  binary: MFTECmd
  sha256: 4b829ec61b34a9817d8eb44ae36e9dfc2e90fdc8f429eec40127ff34cc2f18da
  args: mft_file=/mnt/cases/$MFT, path_filter=None
call-000004 get_amcache [exit=0] 3962 ms 2026-06-14T14:44:14Z
  binary: AmcacheParser
  sha256: bcf639115a6b5995ef56734c76d3d40fd8b758b687a111b2a6f6744d91b9a78d
  args: amcache_hive=/mnt/cases/Windows/appcompat/Programs/Amcache.hve
call-000005 extract_mft_timeline [exit=0] 59051 ms 2026-06-14T14:44:21Z
  binary: MFTECmd
  sha256: 4b829ec61b34a9817d8eb44ae36e9dfc2e90fdc8f429eec40127ff34cc2f18da
  args: mft_file=/mnt/cases/$MFT, path_filter=None
call-000006 get_amcache [exit=0] 6781 ms 2026-06-14T17:32:03Z
  binary: AmcacheParser
  sha256: bcf639115a6b5995ef56734c76d3d40fd8b758b687a111b2a6f6744d91b9a78d
  args: amcache_hive=/mnt/cases/Windows/appcompat/Programs/Amcache.hve
call-000007 extract_mft_timeline [exit=0] 89806 ms 2026-06-14T17:32:26Z
  binary: MFTECmd
  sha256: 4b829ec61b34a9817d8eb44ae36e9dfc2e90fdc8f429eec40127ff34cc2f18da
  args: mft_file=/mnt/cases/$MFT, path_filter=None
call-000008 analyze_prefetch [ERROR] 0 ms 2026-06-14T17:34:38Z
```

args: prefetch\_path=/mnt/cases/Windows/Prefetch/  
error: No .pf files found in /mnt/cases/Windows/Prefetch ? Prefetch may be disabled (common on Windows Server/DC).

call-000009 mem\_pslist [exit=1] 5324 ms 2026-06-14T17:36:01Z

binary: vol

sha256: 9679193c2b7852817006c55481124666422fea67ba63c872cf5e4203c6fa629a

args: memory\_image=/evidence/base-dc-memory.img

call-000010 mem\_netscan [exit=1] 4682 ms 2026-06-14T17:37:34Z

binary: vol

sha256: 9679193c2b7852817006c55481124666422fea67ba63c872cf5e4203c6fa629a

args: memory\_image=/evidence/base-dc-memory.img

call-000011 parse\_event\_logs [exit=-1] 353849 ms 2026-06-14T17:37:53Z

binary: EvtxECmd

sha256: 62c2cb167fc5ab3a77bd7410a04911ecbbf2f345ba094c510565b0d047e1c2e5

args: event\_id=4624, evt\_x\_file=/mnt/cases/Windows/System32/winevt/Logs/Security.evt\_x

call-000012 parse\_event\_logs [exit=-1] 338616 ms 2026-06-14T17:44:05Z

binary: EvtxECmd

sha256: 62c2cb167fc5ab3a77bd7410a04911ecbbf2f345ba094c510565b0d047e1c2e5

args: event\_id=4625, evt\_x\_file=/mnt/cases/Windows/System32/winevt/Logs/Security.evt\_x

call-000013 parse\_event\_logs [exit=0] 64290 ms 2026-06-14T17:50:41Z

binary: EvtxECmd

sha256: 5638bf6e4a63065d631075101456f9696117db6aa08909aaf521de3d33ca6b5f

args: event\_id=7045, evt\_x\_file=/mnt/cases/Windows/System32/winevt/Logs/System.evt\_x

call-000014 extract\_mft\_timeline [exit=0] 100865 ms 2026-06-14T17:53:05Z

binary: MFTECmd

sha256: 4b829ec61b34a9817d8eb44ae36e9dfc2e90fdc8f429eec40127ff34cc2f18da

args: mft\_file=/mnt/cases/\$MFT, path\_filter=Temp

call-000015 extract\_mft\_timeline [exit=0] 81912 ms 2026-06-14T17:54:51Z

binary: MFTECmd

sha256: 4b829ec61b34a9817d8eb44ae36e9dfc2e90fdc8f429eec40127ff34cc2f18da

args: mft\_file=/mnt/cases/\$MFT, path\_filter=rsydow